Customer_Shopping_Behaviour

June 26, 2026

[36]:
```
import pandas as pd
```

[37]:
```
df=pd.read_csv('customer_shopping_behavior.csv')
```

[38]:
```
# DISCOVERY & CLEAN
```

[39]:
```
df.head()
```

[39]:
```
   Customer ID  Age Gender Item Purchased  Category  Purchase Amount (USD)  \
0            1   55   Male         Blouse  Clothing                     53
1            2   19   Male        Sweater  Clothing                     64
2            3   50   Male          Jeans  Clothing                     73
3            4   21   Male        Sandals  Footwear                     90
4            5   45   Male         Blouse  Clothing                     49

        Location Size      Color  Season  Review Rating Subscription Status  \
0       Kentucky    L       Gray  Winter            3.1                 Yes
1          Maine    L     Maroon  Winter            3.1                 Yes
2  Massachusetts    S     Maroon  Spring            3.1                 Yes
3   Rhode Island    M     Maroon  Spring            3.5                 Yes
4         Oregon    M  Turquoise  Spring            2.7                 Yes

  Shipping Type Discount Applied Promo Code Used  Previous Purchases  \
0       Express              Yes             Yes                  14
1       Express              Yes             Yes                   2
2 Free Shipping              Yes             Yes                  23
3  Next Day Air              Yes             Yes                  49
4 Free Shipping              Yes             Yes                  31

  Payment Method Frequency of Purchases
0          Venmo            Fortnightly
1           Cash            Fortnightly
2    Credit Card                 Weekly
3         PayPal                 Weekly
4         PayPal               Annually
```

[40]:
```
df.dtypes
```

[40]:
```
Customer ID                 int64
Age                         int64
Gender                     object
Item Purchased             object
Category                   object
Purchase Amount (USD)       int64
Location                   object
Size                       object
Color                      object
Season                     object
Review Rating             float64
Subscription Status        object
Shipping Type              object
Discount Applied           object
Promo Code Used            object
Previous Purchases          int64
Payment Method             object
Frequency of Purchases     object
dtype: object
```

[41]:
```
#snake_casing format

df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ','_')
```

[42]:
```
df.columns
```

[42]:
```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
       'purchase_amount_(usd)', 'location', 'size', 'color', 'season',
       'review_rating', 'subscription_status', 'shipping_type',
       'discount_applied', 'promo_code_used', 'previous_purchases',
       'payment_method', 'frequency_of_purchases'],
      dtype='object')
```

[43]:
```
df = df.rename(columns={'purchase_amount_(usd)' : 'purchase_amount_usd' })
```

[44]:
```
df.columns
```

[44]:
```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
       'purchase_amount_usd', 'location', 'size', 'color', 'season',
       'review_rating', 'subscription_status', 'shipping_type',
       'discount_applied', 'promo_code_used', 'previous_purchases',
       'payment_method', 'frequency_of_purchases'],
      dtype='object')
```

[45]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   customer_id             3900 non-null   int64
 1   age                     3900 non-null   int64
 2   gender                  3900 non-null   object
 3   item_purchased          3900 non-null   object
 4   category                3900 non-null   object
 5   purchase_amount_usd     3900 non-null   int64
 6   location                3900 non-null   object
 7   size                    3900 non-null   object
 8   color                   3900 non-null   object
 9   season                  3900 non-null   object
 10  review_rating           3863 non-null   float64
 11  subscription_status     3900 non-null   object
 12  shipping_type           3900 non-null   object
 13  discount_applied        3900 non-null   object
 14  promo_code_used         3900 non-null   object
 15  previous_purchases      3900 non-null   int64
 16  payment_method          3900 non-null   object
 17  frequency_of_purchases  3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

[46]:
```
df['review_rating'] = df.groupby("category")['review_rating'].transform(lambda␣
↪x : x.fillna(df['review_rating'].median()))
```

[47]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   customer_id             3900 non-null   int64
 1   age                     3900 non-null   int64
 2   gender                  3900 non-null   object
 3   item_purchased          3900 non-null   object
 4   category                3900 non-null   object
 5   purchase_amount_usd     3900 non-null   int64
 6   location                3900 non-null   object
 7   size                    3900 non-null   object
 8   color                   3900 non-null   object
 9   season                  3900 non-null   object
 10  review_rating           3900 non-null   float64
 11  subscription_status     3900 non-null   object
 12  shipping_type           3900 non-null   object
```

```
 13  discount_applied        3900 non-null   object
 14  promo_code_used         3900 non-null   object
 15  previous_purchases      3900 non-null   int64
 16  payment_method          3900 non-null   object
 17  frequency_of_purchases  3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

[48]:
```
#STRUCTURING
```

[49]:
```
#Label Second Buy vs Others
#============================
import numpy as np

df['purchase_label'] = np.where(
    df['previous_purchases'] < 2,
    'Second Buy',    # Điều kiện True
    'Others'        # Điều kiện False
)

print(df[['previous_purchases', 'purchase_label']].head(8))
```

```
   previous_purchases purchase_label
0                  14         Others
1                   2         Others
2                  23         Others
3                  49         Others
4                  31         Others
5                  14         Others
6                  49         Others
7                  19         Others
```

[57]:
```
# Segmentation following:
# 1. r (frequency of purchase) 2. f (previous purchases) 3. m (purchase amount)␣
 ↪4. rv (review rate)
# ============================================
# BƯỚC 1: RECENCY SCORE - từ frequency_of_purchases
# ============================================

recency_map = {
    'Bi-weekly':       6,
    'Fortnightly':     5,
    'Monthly':         4,
    'Every 3 months':  3,
    'Quarterly':       2,
    'Annually':        1
}
```

```python
df['r_score'] = df['frequency_of_purchases'].map(recency_map)

# ============================================
# BƯỚC 2: FREQUENCY SCORE - từ previous_purchase
# ============================================

df['f_score'] = np.where(
    df['previous_purchases'] < 2,
    1,  # Second Buy → F thấp nhất
    pd.qcut(
        df['previous_purchases'].clip(lower=2),
        q=4,
        labels=[2, 3, 4, 5],
        duplicates='drop'
    ).astype(float)
)

# ============================================
# BƯỚC 3: MONETARY SCORE - từ purchase_amount
# ============================================

df['m_score'] = pd.qcut(
    df['purchase_amount_usd'],
    q=5,
    labels=[1, 2, 3, 4, 5],
    duplicates='drop'
).astype(float)

# ============================================
# BƯỚC 4: REVIEW SCORE - từ review_rate
# ============================================

df['review_score'] = pd.cut(
    df['review_rating'],
    bins=[0, 3, 4, 5],
    labels=[1, 2, 3]   # 1-3 → Thấp | 3-4 → Trung bình | 4-5 → Cao
).astype(float)

# ============================================
# BƯỚC 5: TỔNG HỢP RFM + REVIEW
# ============================================

df['rfm_score'] = (
    df['r_score']      * 0.25 +
    df['f_score']      * 0.30 +
    df['m_score']      * 0.30 +
    df['review_score'] * 0.15
```

```
).round(2)

# ============================================
# BƯỚC 6: GÁN SEGMENT
# ============================================

def assign_segment(row):
    r = row['r_score']
    f = row['f_score']
    m = row['m_score']
    rv = row['review_score']

    #
    # GUARD CLAUSE - Kiểm tra null trước
    # Bất kỳ score nào null → không đủ data
    # → Xếp vào Reconsider, không tính RFM
    #
    if pd.isna(r) or pd.isna(f) or pd.isna(m) or pd.isna(rv):
        return 'Reconsider'

    #
    # 1. CHAMPIONS - Khách hàng VIP
    # R, F, M đều cao nhất + hài lòng cao
    #
    if r >= 4 and f >= 4 and m >= 4 and rv >= 3:
        return 'Champions'

    #
    # 2. LOYAL CUSTOMERS - Khách hàng trung thành
    # F và M cao, quay lại thường xuyên + hài lòng
    # Không nhất thiết vừa mua gần đây
    #
    elif f >= 4 and m >= 3 and rv >= 3:
        return 'Loyal Customers'

    #
    # 3. POTENTIAL LOYALIST - Khách hàng tiềm năng
    # Mới mua gần đây, tần suất bắt đầu đều
    # Chi tiêu khá, review trung bình trở lên
    #
    elif r >= 4 and f >= 2 and m >= 3 and rv >= 2:
        return 'Potential Loyalist'

    #
    # 4. NEW CUSTOMERS - Khách hàng mới
    # R rất cao (vừa mua gần đây)
    # F và M còn thấp, review chưa có nhiều
```

```python
    # → Không filter rv vì khách mới chưa đủ data đánh giá
    #
    elif r >= 4 and f <= 2 and m <= 2:
        return 'New Customers'

    #
    # 5. PROMISING - Khách hàng hứa hẹn
    # Tần suất ổn, đang tăng dần
    # Giá trị chưa cao nhưng review tích cực
    #
    elif r >= 3 and f >= 3 and m <= 3 and rv >= 2:
        return 'Promising'

    #
    # 6. NEEDS ATTENTION - Khách cần quan tâm
    # Tất cả chỉ số ở mức trung bình
    # rv trung bình → chưa đủ hài lòng để gắn bó lâu dài
    #
    elif r >= 2 and f >= 2 and m >= 2 and rv <= 1:
        return 'Needs Attention'

    #
    # 7. CAN'T LOSE THEM - Không thể mất
    # Từng mua rất nhiều & thường xuyên (F, M cao)
    # R thấp - đang rời đi
    # rv thấp → không hài lòng là lý do rời đi
    #
    elif r <= 2 and f >= 4 and m >= 4 and rv <= 1:
        return "Can't Lose Them"

    #
    # 8. AT RISK - Nguy cơ mất khách
    # Từng mua khá nhưng lâu rồi chưa quay lại
    # rv thấp → có thể là lý do giảm tương tác
    #
    elif r <= 2 and (f >= 3 or m >= 3) and rv <= 1:
        return 'At Risk'

    #
    # 9. HIBERNATING / LOST - Khách rời bỏ
    # R, F, M đều thấp + review tệ
    # Lâu không tương tác, gần như mất hẳn
    #
    else:
        return 'Hibernating/Lost'
```

```python
df['segment'] = df.apply(assign_segment, axis=1)

# ============================================
# BƯỚC 7: KIỂM TRA KẾT QUẢ
# ============================================

# Phân phối segment
print("=== PHÂN PHỐI SEGMENT ===")
print(df['segment'].value_counts())
print(f"\nTổng khách: {len(df)}")

# Profile trung bình từng segment
print("\n=== PROFILE TRUNG BÌNH TỪNG SEGMENT ===")
profile = df.groupby('segment').agg(
    so_luong_khach   = ('segment',          'count'),
    r_score_tb       = ('r_score',          'mean'),
    f_score_tb       = ('f_score',          'mean'),
    m_score_tb       = ('m_score',          'mean'),
    review_rating_tb  = ('review_rating',    'mean'),
    rfm_score_tb     = ('rfm_score',        'mean'),
    purchase_amt_tb  = ('purchase_amount_usd', 'mean'),
    prev_purchase_tb = ('previous_purchases','mean')
).round(2)

print(profile)

# ============================================
# BƯỚC 8: EXPORT KẾT QUẢ
# ============================================

cols_export = [
    'customer_id',
    'frequency_of_purchases',
    'previous_purchases',
    'purchase_amount_usd',
    'review_rating',
    'r_score',
    'f_score',
    'm_score',
    'review_rating',
    'rfm_score',
    'segment'
]

df[cols_export].to_csv('rfm_segments.csv', index=False)
print("\n  Đã export file rfm_segments.csv")
```

```
=== PHÂN PHỐI SEGMENT ===
segment
Reconsider            1670
Hibernating/Lost       836
Potential Loyalist     373
Needs Attention        260
Promising              257
Loyal Customers        181
New Customers          115
At Risk                112
Champions               80
Can't Lose Them         16
Name: count, dtype: int64

Tổng khách: 3900

=== PROFILE TRUNG BÌNH TỪNG SEGMENT ===
                    so_luong_khach  r_score_tb  f_score_tb  m_score_tb  \
segment
At Risk                        112        1.23        3.58        2.29
Can't Lose Them                 16        1.00        4.50        4.56
Champions                       80        4.47        4.46        4.49
Hibernating/Lost               836        1.69        3.28        2.77
Loyal Customers                181        2.28        4.46        3.77
Needs Attention                260        3.62        3.52        3.52
New Customers                  115        4.59        1.91        1.50
Potential Loyalist             373        4.47        3.19        3.95
Promising                      257        4.47        3.96        1.48
Reconsider                    1670         NaN        3.44        2.99

                    review_rating_tb  rfm_score_tb  purchase_amt_tb  \
segment
At Risk                         2.77          2.22            49.07
Can't Lose Them                 2.79          3.12            86.06
Champions                       4.54          4.25            83.50
Hibernating/Lost                3.82          2.57            56.06
Loyal Customers                 4.56          3.49            72.91
Needs Attention                 2.77          3.17            68.70
New Customers                   3.74          2.49            35.64
Potential Loyalist              3.89          3.61            75.82
Promising                       4.02          3.12            35.61
Reconsider                      3.75           NaN            59.92

                    prev_purchase_tb
segment
At Risk                        27.01
Can't Lose Them                38.81
Champions                      37.64
```

```
Hibernating/Lost             23.24
Loyal Customers              37.58
Needs Attention              26.25
New Customers                 6.86
Potential Loyalist           22.06
Promising                    31.58
Reconsider                   25.17
```

```
 Đã export file rfm_segments.csv
```

[51]:
```python
pd.set_option("display.max_column", None)
df
```

[51]:
```
      customer_id  age  gender item_purchased     category  \
0               1   55    Male         Blouse     Clothing
1               2   19    Male        Sweater     Clothing
2               3   50    Male          Jeans     Clothing
3               4   21    Male        Sandals     Footwear
4               5   45    Male         Blouse     Clothing
…             …   …      …             …            …
3895         3896   40  Female         Hoodie     Clothing
3896         3897   52  Female       Backpack  Accessories
3897         3898   46  Female           Belt  Accessories
3898         3899   44  Female          Shoes     Footwear
3899         3900   52  Female        Handbag  Accessories

      purchase_amount_usd       location size      color  season  \
0                      53       Kentucky    L       Gray  Winter
1                      64          Maine    L     Maroon  Winter
2                      73  Massachusetts    S     Maroon  Spring
3                      90   Rhode Island    M     Maroon  Spring
4                      49         Oregon    M  Turquoise  Spring
…                     …            …  …    …         …
3895                   28       Virginia    L  Turquoise  Summer
3896                   49           Iowa    L      White  Spring
3897                   33     New Jersey    L      Green  Spring
3898                   77      Minnesota    S      Brown  Summer
3899                   81     California    M      Beige  Spring

      review_rating subscription_status   shipping_type discount_applied  \
0               3.1                 Yes         Express              Yes
1               3.1                 Yes         Express              Yes
2               3.1                 Yes   Free Shipping              Yes
3               3.5                 Yes    Next Day Air              Yes
4               2.7                 Yes   Free Shipping              Yes
…               …                   …             …                …
3895            4.2                  No  2-Day Shipping               No
```

```
3896           4.5                  No   Store Pickup                No
3897           2.9                  No       Standard                No
3898           3.8                  No        Express                No
3899           3.1                  No   Store Pickup                No

     promo_code_used  previous_purchases payment_method  \
0                Yes                  14          Venmo
1                Yes                   2           Cash
2                Yes                  23    Credit Card
3                Yes                  49         PayPal
4                Yes                  31         PayPal
…               …                  …            …
3895              No                  32          Venmo
3896              No                  41  Bank Transfer
3897              No                  24          Venmo
3898              No                  24          Venmo
3899              No                  33          Venmo

     frequency_of_purchases purchase_label  r_score  f_score  m_score  \
0               Fortnightly         Others      5.0      3.0      3.0
1               Fortnightly         Others      5.0      2.0      3.0
2                    Weekly         Others      NaN      3.0      4.0
3                    Weekly         Others      NaN      5.0      5.0
4                  Annually         Others      1.0      4.0      2.0
…                      …             …      …      …      …
3895                 Weekly         Others      NaN      4.0      1.0
3896              Bi-Weekly         Others      NaN      5.0      2.0
3897              Quarterly         Others      2.0      3.0      1.0
3898                 Weekly         Others      NaN      3.0      4.0
3899              Quarterly         Others      2.0      4.0      4.0

      review_score  rfm_score             segment
0              2.0       3.35  Potential Loyalist
1              2.0       3.05  Potential Loyalist
2              2.0        NaN          Reconsider
3              2.0        NaN          Reconsider
4              1.0       2.20             At Risk
…             …          …                 …
3895           3.0        NaN          Reconsider
3896           3.0        NaN          Reconsider
3897           1.0       1.85             At Risk
3898           2.0        NaN          Reconsider
3899           2.0       3.20    Hibernating/Lost

[3900 rows x 25 columns]
```

[52]:
```
age_labels = ["Young adult", "adult", "Middle-aged", "Senior"]

# Tạo cột mới 'Age_Group' bằng pd.cut
df['age_group'] = pd.cut(df['age'], bins=4, labels=age_labels)
```

[53]:
```
df.columns
```

[53]:
```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
       'purchase_amount_usd', 'location', 'size', 'color', 'season',
       'review_rating', 'subscription_status', 'shipping_type',
       'discount_applied', 'promo_code_used', 'previous_purchases',
       'payment_method', 'frequency_of_purchases', 'purchase_label', 'r_score',
       'f_score', 'm_score', 'review_score', 'rfm_score', 'segment',
       'age_group'],
      dtype='object')
```

[54]:
```
df.describe(include='all')
```

[54]:
```
        customer_id          age gender item_purchased  category  \
count   3900.000000  3900.000000   3900           3900      3900
unique          NaN          NaN      2             25         4
top             NaN          NaN   Male         Blouse  Clothing
freq            NaN          NaN   2652            171      1737
mean    1950.500000    44.068462    NaN            NaN       NaN
std     1125.977353    15.207589    NaN            NaN       NaN
min        1.000000    18.000000    NaN            NaN       NaN
25%      975.750000    31.000000    NaN            NaN       NaN
50%     1950.500000    44.000000    NaN            NaN       NaN
75%     2925.250000    57.000000    NaN            NaN       NaN
max     3900.000000    70.000000    NaN            NaN       NaN

        purchase_amount_usd location  size  color  season  review_rating  \
count           3900.000000     3900  3900   3900    3900    3900.000000
unique                  NaN       50     4     25       4            NaN
top                     NaN  Montana     M  Olive  Spring            NaN
freq                    NaN       96  1755    177     999            NaN
mean              59.764359      NaN   NaN    NaN     NaN       3.750538
std               23.685392      NaN   NaN    NaN     NaN       0.713589
min               20.000000      NaN   NaN    NaN     NaN       2.500000
25%               39.000000      NaN   NaN    NaN     NaN       3.100000
50%               60.000000      NaN   NaN    NaN     NaN       3.800000
75%               81.000000      NaN   NaN    NaN     NaN       4.400000
max              100.000000      NaN   NaN    NaN     NaN       5.000000

       subscription_status  shipping_type discount_applied promo_code_used  \
count                 3900           3900             3900            3900
unique                   2              6                2               2
```

```
top                     No  Free Shipping              No             No
freq                  2847            675            2223           2223
mean                   NaN            NaN             NaN            NaN
std                    NaN            NaN             NaN            NaN
min                    NaN            NaN             NaN            NaN
25%                    NaN            NaN             NaN            NaN
50%                    NaN            NaN             NaN            NaN
75%                    NaN            NaN             NaN            NaN
max                    NaN            NaN             NaN            NaN

        previous_purchases payment_method frequency_of_purchases  \
count          3900.000000           3900                   3900
unique                 NaN              6                      7
top                    NaN         PayPal         Every 3 Months
freq                   NaN            677                    584
mean             25.351538            NaN                    NaN
std              14.447125            NaN                    NaN
min               1.000000            NaN                    NaN
25%              13.000000            NaN                    NaN
50%              25.000000            NaN                    NaN
75%              38.000000            NaN                    NaN
max              50.000000            NaN                    NaN

       purchase_label      r_score      f_score      m_score  review_score  \
count            3900  2230.000000  3900.000000  3900.000000   3900.000000
unique              2          NaN          NaN          NaN           NaN
top            Others          NaN          NaN          NaN           NaN
freq             3817          NaN          NaN          NaN           NaN
mean              NaN     2.968610     3.452821     2.980000      2.156667
std               NaN     1.580756     1.152884     1.419503      0.751040
min               NaN     1.000000     1.000000     1.000000      1.000000
25%               NaN     1.000000     2.000000     2.000000      2.000000
50%               NaN     2.000000     3.000000     3.000000      2.000000
75%               NaN     4.000000     4.000000     4.000000      3.000000
max               NaN     5.000000     5.000000     5.000000      3.000000

          rfm_score     segment    age_group
count   2230.000000        3900         3900
unique          NaN          10            4
top             NaN  Reconsider  Young adult
freq            NaN        1670         1028
mean       2.996256         NaN          NaN
std        0.680606         NaN          NaN
min        1.000000         NaN          NaN
25%        2.500000         NaN          NaN
50%        2.950000         NaN          NaN
75%        3.500000         NaN          NaN
```

```
max        4.700000         NaN          NaN
```

[55]:
```
!pip install sqlalchemy psycopg2
```

```
Requirement already satisfied: sqlalchemy in
c:\users\admin\appdata\local\programs\python\python313\lib\site-packages
(2.0.50)
Requirement already satisfied: psycopg2 in
c:\users\admin\appdata\local\programs\python\python313\lib\site-packages
(2.9.12)
Requirement already satisfied: greenlet>=1 in
c:\users\admin\appdata\local\programs\python\python313\lib\site-packages (from
sqlalchemy) (3.5.1)
Requirement already satisfied: typing-extensions>=4.6.0 in
c:\users\admin\appdata\local\programs\python\python313\lib\site-packages (from
sqlalchemy) (4.12.2)


[notice] A new release of pip is available: 25.3 -> 26.1.2
[notice] To update, run: python.exe -m pip install --upgrade pip
```

[56]:
```
from sqlalchemy import create_engine

username = 'postgres'
password = 'Highlands'
host = 'localhost'
post = '5432'
database = 'Customer_Shopping_Behavior'

engine = create_engine('postgresql+psycopg2://postgres:Highlands@localhost:5432/
 ↪Customer_Shopping_Behavior')

# 3. Đẩy toàn bộ dữ liệu vào Database chỉ với 1 dòng lệnh!
try:
    # if_exists='replace': Nếu bảng đã có, nó sẽ xóa đi và tạo lại bản chuẩn␣
 ↪xác nhất
    # index=False: Không đưa cột số thứ tự mặc định của Python vào Database
    df.to_sql('shopping_event', engine, if_exists='replace', index=False)
    print("  Import dữ liệu thành công rực rỡ!")
except Exception as e:
    print("Có lỗi xảy ra:", e)
```

```
  Import dữ liệu thành công rực rỡ!
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